

Sensor-Based Human Activity Recognition: A Comparative Study of Machine Learning Techniques

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Abstract—Human activity recognition (HAR) is a wide field of study which identifies a person's specific movement or behavior based on sensor data. Recognition of human behavior is the origin of many technologies, such as those concerned with personal biometric signatures, sports training, digital computing, security, health and fitness tracking, ambient-assisted living and management. Studying recognition of human activity shows that researchers are mostly interested in human everyday activities. HAR models input is the reading of the raw sensor data, and output is the prediction of the movement activities of the user. The HAR framework is becoming an evolving discipline in intelligent computing applications in the field of pervasive computing. In our study, we applied several machine learning algorithm along with some preprocessing techniques to identify which algorithm performs better in dataset acquired from the WISDM laboratory, which is available in public domain. The experiment shows that the highest accuracy is achieved in phone accelerometer data using Principal Component Analysis (PCA) with Random Forest (RF) than any other algorithm and preprocessing techniques in terms of human activity recognition. This experiment will help perform more work on the basis of implementing classification and preprocessing techniques to identify human activities.

Index Terms—Human activity recognition, machine Learning, preprocessing, accelerometer, gyroscope, accuracy

I. INTRODUCTION

Human activity recognition is a process in which a subject's movements are read and predict the physical activity of the subject. Most of this type of research has started to use commercial smartphones and smartwatches in recent years, including the required sensors like accelerometers and gyroscopes [1] [2]. The accelerometer and gyroscope have proved useful in capturing required movements and understanding physical activities of humans [3]. The sensors of these kinds of smart devices are also useful because of the high availability and frequent use of the subject's in daily life. HAR has become a promising topic of research in the areas of healthcare systems [4], robotic systems [5], smart home systems [6] and many others.

But there are still some challenges ahead. Smartphone sensors face some problems for female subjects in terms of some activity including walking and running as they typically carry mobile phones on their purse, not pockets. Smartwatch

sensors face problems with some activities, such as eating, using cell phones or computers and others. It occurs due to some people wearing watches in their left hand and some wearing in their right. And some are left handed and some are right handed. In this research, these kinds of challenges cause some lake of performance.

HAR is a very challenging problem when the prediction is made using sensor data from the smart device. The style of performing the same physical activity varies from person to person. In general, HAR can be treated as a pattern recognition issue through smart device sensors. Each and every physical activity surely has some very common physical movements. Like a pattern recognition technique, a HAR model attempts to analyze and marginalize the sensor data, find out the co-relations and predict the activity.

In this research, Machine learning algorithms are used before and after some preprocessing of data on each dataset. The aim was to find out which algorithm performs better after different preprocessing method. The different datasets also exhibited different behaviors at each point. As a findings of this study, these results and several other important points are discussed in result analysis. Most of the other related researches listed here focused mainly on the algorithm aspect to find out the best possible accuracy. But the preprocessing methods are also getting enough attention in this research so are the algorithms. And it makes a major impact on the calculation of accuracy.

Rest of the paper is organized as follows. In section II, the state of the art for HAR based on sensor data is described. In section III, the methodology and procedure is described briefly. In section IV, the findings and outcomes of this research is described. Finally the research is concluded in section V with the outline of the future work.

II. RELATED WORK

In recent years several experiments were performed for identification of human activity recognition. In [7], the authors look at how smartphones and smartwatches are used to identify human activity and data were gathered from 17 test subjects for this study. The findings indicate that the watch accelerometer produces much better results than the phone accelerometer

with an overall accuracy of 91.9% versus 72.6% for personal models and 64.0% versus 25.3% for impersonal models.

The authors in [8] provide an inclusive assessment of Biometrics smartphones dependent on motion. This study consists of 51 test subjects spending three minutes doing 18 activities. The authors implemented KNN, RF and Decision Tree (DT) for performance measurement of sensor combination. The authors found that the accelerometer combination gives overall best results than others and in terms of the accelerometer combination Random forest gives better accuracy 99.7% with voting and 96.4% without voting.

In [9], the authors explored the recognition of complex activities with common classifiers. Data was collected about the subject doing breakfast scenarios. The activities are classified into three level hierarchy so that the complex activities in the high level and the simple in the low level. Logistic Regression (LR), NB, C4.5, Neural Network (NN), Support Vector Machine (SVM), KNN and DT were applied for accuracy measurement. The result showed that low level activity gives the highest accuracy 91.05% got by KNN.

In [10], the authors applied a multichannel CNN-based model to classify the human activity classification and test it on the UCI HAR dataset to extract frequency and power information from the accelerometer's raw time domain signal. The authors obtained 95.25% classification accuracy.

In [11], the author applied Decision Stump, Random Tree, Hoeffding Tree, REP Tree, J48 and Random Forest, decision tree algorithms along with Adaptive Boosting to classify the human activity on standard HAR dataset. As a result, the authors found that the accuracy of Random forest, REP Tree and J48 compared to Decision Stump, and Hoeffding Tree is statistically meaningful.

In [12], the authors conducted a study by implementing the K-nearest Neighbor algorithm to classify 6 different human activity recognition. Data was collected using an android smartphone with a built-in accelerometer sensor and MATLAB is used to process those collected data. KNN was applied using WEKA analysis software with k value of 3 and an accuracy of 97.97% was achieved.

The authors in [13] developed a system to identify six ordinary human actions using spatial information obtained from Ubisense positioning system. The system is implemented using convolutional neural networks (CNN). The authors achieved a best accuracy of 86.7% from different experiments.

In [14], the authors proposed a classification system for human action recognition system on UFC50 dataset using genetic algorithm and CNN. Authors demonstrate that it minimizes the classification error by initializing the weights of a CNN classifier based on solutions created by genetic algorithms. The proposed approach achieves a recognition accuracy of 99.98%.

III. METHODOLOGY

We followed several steps as shown in the figure 1. At first, we gathered the dataset and then used some preprocessing techniques to transform the raw data in a useful and efficient format. In this preprocessing stage we have used two types

of feature scaling method such as Standardization and Normalization and also used linear transformation techniques like Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA). We applied these preprocessing methods in all features of the datasets. After that we applied some classification method in the transformed dataset as well as in the raw dataset. Lastly, we calculated the accuracy of those classification method for further analysis.

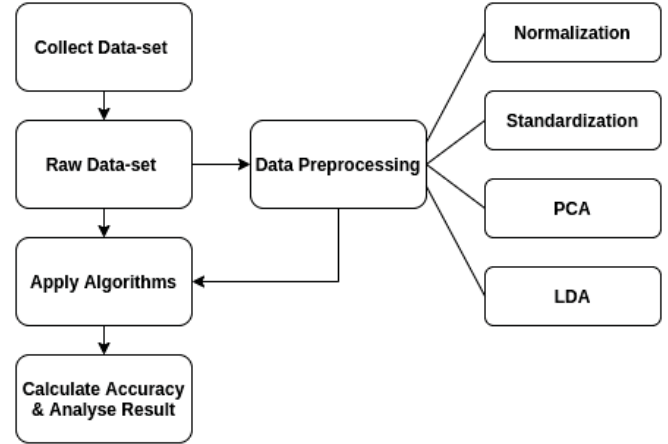


Fig. 1: Flow Chart of Proposed Methodology

A. Dataset Description

Wireless Sensor Data Mining (WISDM) Lab gathers the data set used in this study [15]. They mainly concentrate on collecting sensor data from smartphones and other modern devices such as tablets, smart watches etc. and trying to extract those data for further studies. The dataset used here is named "WISDM Smartphone and Smartwatch Activity and Biometrics Dataset". Here 51 subjects are volunteered to perform for 18 separate physical activities, 3 minutes for each. Volunteers kept a smartphone in their preferred pocket and held a smart-watch in their preferred hand while the sensor reading were noted.

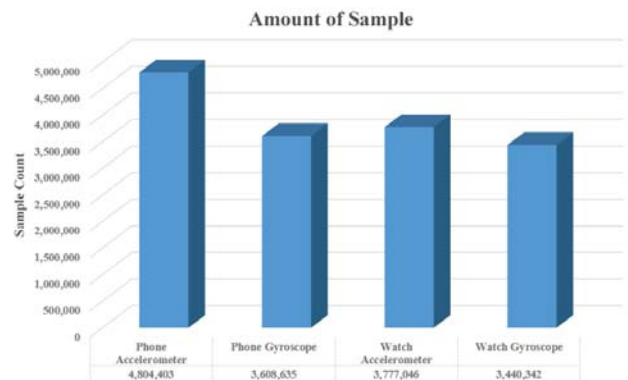


Fig. 2: Overview of Datasets

In both the phone and watch at a rate of 20Hz, the raw sensor data is reported by the accelerometer and the gyroscope.

Data was stored in separate sub-directories for each of the four sensors. In addition, there are 51 separate files for each subject within the sub-directories. Each data set line is structured as follows: Subject-id, Activity Code, Timestamp, x , y, z. The dataset includes 15,630,426 individual sensor readings across four sub-directories. But when collecting the data, there was some failure and four sub-directories don't contain the same amount of data for this (Figure 2). The gyroscopes provided fewer data for smartphone as well as smartwatch. When the watch gyroscopes provided the least amount of data, the phone accelerometers provided the most data.

B. Data Preprocessing

We used two types of feature scaler, first one is Standardization and another one is Normalization. Sometimes, datasets might have many types of attributes and each and every attribute might not be in the same range that means one attribute's values might be higher and another might be lower which might influence the model accuracy. For that reason, Standardization and Normalization are the values into a fixed range which have a big impact on model accuracy in machine learning.

Standardization is a technique where the values are centered around the mean with a unit standard deviation. This means that the distribution centered around 0, with a standard deviation of 1. For the Standardization, we computed both standard deviation and mean of each and every feature which is adjusted by minus mean and divided by standard deviation.

Normalized equation is formed by initially subtracting the minimum value from the variable divided by the minimum value is Subtract from maximum value which means all independence variables are reduced and produced a fixed range.

Principal Component Analysis (PCA) is an unsupervised technique which is used for dimensionality reduction. PCA also can reduce noise in datasets. After using PCA, datasets can be easily visualized. Reduced dataset has an impact on the result. For the reduction dimension, many algorithms give best accuracy and many of them give nearly accurate results.

Linear discriminant analysis (LDA) is a supervised technique when a dataset works for classification and segmented analysis. It also reduced dimensions like PCA. LDA makes assumptions for normally distributed classes and equal class covariances. LDA reduces the number of features and makes a more manageable number before the process of classification. LDA generates a new linear combination of features which helps to enhance the algorithm's accuracy.

C. Classifiers

Random Forest (RF) is a supervised algorithm and it is a modified version of decision trees. It consists of a large number of individual decision trees which operate as an ensemble. Each and every individual tree split out a class prediction in random forest and selected the predictive class using a voting system.

K-Nearest Neighbors (KNN) used for both classification and regression problems. It classifies new data points using the similarity measures and the classification done by major votes to its neighbors that means it votes the most frequent labels for classification problems and takes the average label for regression problems.

Long Short-Term Memory (LSTM) is a deep learning algorithm which is a type of recurrent neural network. LSTM has capabilities to complex domains such as speech recognition, machine translation and also worked on time sequential data. LSTM also has capabilities of sequence prediction problems based on order dependencies.

Support Vector Machine (SVM) is a supervised algorithm in machine learning, it can be used for both classification and regression challenges. SVM also classifies linear or non-linear problems and it separates the data into multiple classes using a line or a hyperplane.

Multilayer Perceptron (MLP) classifier connects to a neural network. MLP is a class of feedforward artificial neural network (ANN) and it utilizes a supervised technique which is called backpropagation for training data.

IV. RESULTS AND EXPERIMENTS

A. Implementation

In order to implement our method we have used anaconda which is an open-source distribution of python programming language. Anaconda is the most popular tool for data science because it includes various libraries of several well-known algorithms. The other configurations are:

Processor: Intel Core i5, Clock Rate: 2.3GHz, RAM: 8GB, Operating System: Windows 10

B. Accuracy Comparison

As we discussed before, researchers used several algorithms to identify human activities. In this research We have applied several algorithms along with several preprocessing techniques to get better accuracy of each dataset. And since the phone and watch both have an accelerometer and gyroscope sensor, we also consider the combination of phone and watch accelerometer data as well as phone and watch gyroscope data in our experiment. We used RF, KNN, LSTM, SVM and MLP in this study. While applying these algorithms we randomly choose 75% sample to train our model and 25% sample to test our model. And the calculated results are average accuracy of 18 activities. Now we will discuss about the accuracy of these algorithms with some preprocessing techniques.

TABLE I: Accuracy(%) using Raw Dataset

Algorithms	Phone		Watch		Phone+Watch	
	Accel	Gyro	Accel	Gyro	Accel	Gyro
RF	86.35	57.82	70.9	70.9	81.29	63.29
K-NN	59.65	24.12	36.0	36.0	49.24	29.14
LSTM	77.24	44.28	66.70	66.52	68.65	52.07
SVM	60.59	29.62	45.8	45.8	54.81	36.97
MLPClassifier	72.7	41.83	63.8	63.8	65.17	51.84

TABLE II: Accuracy(%) using Feature Scaling

Algorithms	Phone				Watch				Phone + Watch			
	Accel		Gyro		Accel		Gyro		Accel		Gyro	
	Norm.	Std.	Norm.	Std.	Norm.	Std.	Norm.	Std.	Norm.	Std.	Norm.	Std.
RF	86.4	86.4	57.5	57.6	70.0	70.0	70.0	70.0	80.7	80.6	63.0	62.8
K-NN	81.9	83.8	53.4	56.9	62.6	65.5	62.6	65.5	71.8	75.9	56.1	60.3
LSTM	84.7	79.5	45.5	55.91	68.21	73.72	69.85	73.02	70.02	77.68	55.8	61.76
SVM	37.1	68.5	21.7	40.0	44.9	63.1	44.9	63.1	34.6	63.0	26.8	50.2
MLPClassifier	72.9	79.9	41.6	48.2	66.0	69.3	66.0	69.3	66.7	71.9	52.9	56.2

TABLE III: Accuracy(%) using PCA and LDA

Algorithms	Phone				Watch				Phone+Watch			
	Accel.		Gyro		Accel		Gyro		Accel		Gyro	
	PCA	LDA	PCA	LDA	PCA	LDA	PCA	LDA	PCA	LDA	PCA	LDA
RF	87.5	83.2	59.3	55.1	71.7	70.3	71.7	70.3	81.20	76.4	64.4	60.2
K-NN	58.2	83.6	24.7	56.7	35.5	68.7	35.5	68.7	48.5	76.6	29.0	59.9
LSTM	83.72	82.14	54.8	55.0	74.23	73.14	73.54	71.69	76.82	74.51	60.86	58.9
SVM	59.6	74.6	29.3	49.4	46.2	71.0	46.2	71.0	54.1	69.5	37.0	56.4
MLPClassifier	78.2	72.4	47.3	47.5	70.3	68.7	70.3	68.7	70.9	63.7	57.0	52.4

1) *Raw Dataset*: At first, we have applied the algorithms in the raw datasets. The accuracy of several algorithm are depicted in table 1,

From table 1, we can see that RF gives the best accuracy in both Phone Accelerometer (Accel) (86.3%) and Gyroscope (Gyro) (57.8%). For watch data, again RF gives the best accuracy in both Accelerometer (70.9%) and Gyroscope (70.9%). For combine phone and watch accelerometer data, Again RF gives the best accuracy(81.29%). And for combine phone and watch gyroscope data, also RF gives the best accuracy(63.29%) than others. So, overall RF gives the best accuracy among raw datasets and the highest accuracy (86.3%) achieved in phone accelerometer data.

2) *Feature Scaling*: Feature scaling has some impact on the performance of some algorithms such as K-NN, LSTM, MLPClassifier in accuracy improvement¹. Two most widely recognized strategies of feature scaling are Normalization and Standardization².

Several algorithms accuracy after using feature scaling are shown in table 2,

From table 2, we can see that after applying normalization and standardization on phone accelerometer data, RF gives the best accuracy which is 86.4% but the accuracy is almost same as the one from table 1. Also, we can see that other algorithm has some accuracy improvement compared to raw data accuracy. RF also gives the best accuracy (57.6%) When we applied feature scaling on phone gyroscope data but the accuracy is also almost same as the one from table 1. So, we can say that after using feature scaling, K-NN, LSTM, SVM, MLPClassifier have some accuracy improvement but these accuracies are less compared to RF.

Now if we see the accuracy of watch accelerometer and gyroscope data, when we use normalization in watch accelerometer and gyroscope data, RF gives the best accuracy which is as same as raw data accuracy. But after applying

standardization, LSTM gives the best accuracy for both watch accelerometer (73.7%) and gyroscope (73%). So, after applying standardization technique, LSTM gives best accuracy than the raw data accuracy.

For combine phone and watch accelerometer data, we see that RF gives best accuracy and the accuracies are almost same for both normalization(80.7%) and standardization(80.6%). For combine phone and watch gyroscope data, Random Forest gives best accuracy than others and which is almost same for normalization(63%), standardization(62.8%).

So, after applying feature scaling, LSTM outperformed RF in both watch accelerometer and gyroscope data compared to raw data accuracy but the highest accuracy (86.4%) achieved in phone accelerometer data got by RF.

3) *PCA and LDA*: The reason of using PCA and LDA was because it gives a lower dimension of the dataset which can affect the outcome [16]. Now several algorithms accuracy using PCA and LDA is given in table 3,

From table 3, we see that RF algorithm gives best accuracy (87.5%) in phone accelerometer data after using PCA which is higher than raw dataset and feature scaling accuracy. And also phone gyroscope accuracy is higher than before, when we use PCA and RF algorithm (59.3%).

Now for watch accelerometer, we can see that PCA with LSTM gives the best accuracy (74.2%) than before. And also for watch gyroscope LSTM with PCA gives the best accuracy (73.5%), which is better accuracy than raw dataset and feature scaling accuracy.

For combine phone and watch accelerometer data, we see that RF with PCA gives best accuracy(81.20%) than other algorithms. And for combine phone and watch gyroscope data, RF with PCA gives better accuracy(64.4%) than other algorithms.

So, after applying PCA and LDA, LSTM gives better accuracy than RF in both watch accelerometer and gyroscope data compared to raw data accuracy but overall RF gives the highest accuracy (87.5%) which is achieved in phone accelerometer dataset.

¹<https://becominghuman.ai/demystifying-feature-scaling>

²<https://towardsdatascience.com/all-about-feature-scaling>

So From our study we can say that PCA with RF gives the best accuracy in phone accelerometer and gyroscope dataset and PCA with LSTM gives best accuracy in watch accelerometer and gyroscope dataset. For combine phone and watch accelerometer dataset, RF gives best accuracy in raw dataset and for combine phone and watch gyroscope dataset, PCA with RF gives best accuracy. Among all the results before and after applying preprocessing techniques, PCA with RF performs best to recognize human activity in phone accelerometer dataset.

C. Comparison With Other Studies

Human activity recognition is one of the challenging part in machine learning. Many researchers are working to recognize various human activity. There are different types of analysis on human activity recognition, so it's quite difficult to compare with others.

Weiss et al. used various algorithm such as RF, NB, MLPClassifier etc. in the WISDM dataset and proved that smartwatch can more capable than smartphone to identify hand-based activities, but they just used 17 subjects [7]. But our study consists of 51 subjects and we used many preprocessing techniques to improve the accuracy.

Yoneda et al. used their own model named mobile biometric models to recognize human activity and try to find accuracy of single activity using RF [8]. In our study, we used several algorithms with various preprocessing techniques to improve accuracy of human activity recognition.

V. CONCLUSION AND FUTURE WORK

Human activity recognition which is predicting the movement of a person based on sensor data, a challenging task in machine learning³. In our study, we have applied various algorithms to identify the best algorithm. We have also applied different preprocessing techniques to improve the accuracy. We found that PCA with RF algorithm gives the best accuracy regarding phone accelerometer (87.5%) and gyroscope (59.3%) data. For watch accelerometer and gyroscope data, PCA with LSTM gives the best result of 74.2% and 73.5%. For combine phone and watch accelerometer data RF gives best accuracy (81.29%) in raw data and for combine phone and watch gyroscope data, RF gives best accuracy (64.4%) after applying PCA. And among all the accuray, the highest accuray (87.5%) obtained in phone acclerometer dataset after appying PCA and RF.

The research mentioned in this paper represents a significant phase in our strategy. We expect this analysis to be carried out in a number of ways. We will be trying to build up an application for smartphones for the secondary level. It will be a health monitoring program that will produce a brief overview of the day-to-day operation of the subject and provide relevant health advice.

³<https://machinelearningmastery.com/deep-learning-models-for-human-activity-recognition>

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